Holography with Physical optics

1 Holography introduction

Antenna holography is procedure with the goal of measuring the antenna positioning imperfections to later correct them.

The ideal cassegrain antenna is composed by a big main dish with a paraboloidal form and a subreflector with an hyperboloid geometry. While the subreflector can be build with just a single metallic piece, the main parabolic dish is normally build placing several panels that composed the desired geometry. In figure 1 you can see the APEX diagram as example.

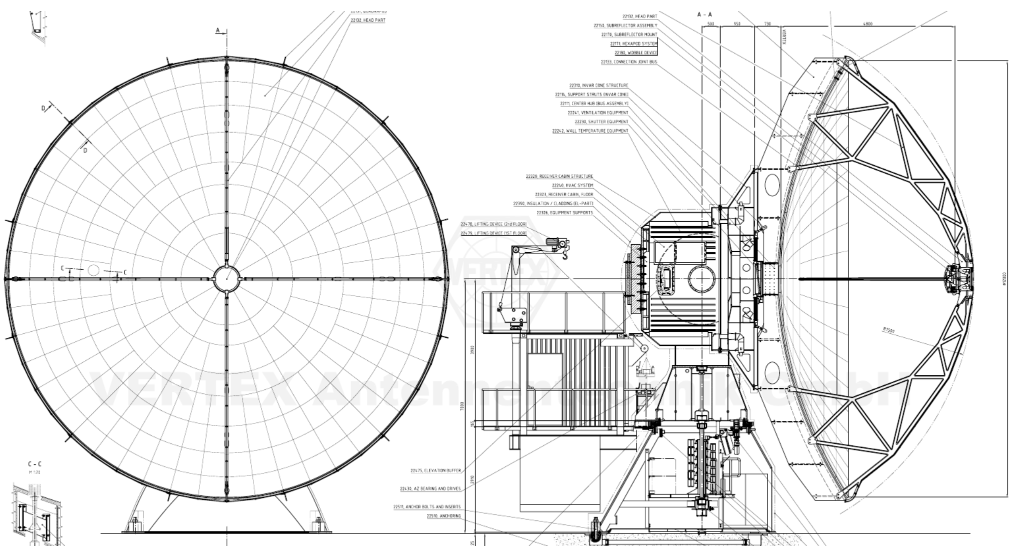

Figure 1: APEX telescope diagram.

The accuracy of the panel positioning affects directly the efficiency of the overall antenna, you can think of it as that as worst the surface of the antenna you are not collecting all the power in the foci. The rule of thumbs is that the main dish should have an error in the positioning of the panels of around $\lambda/16$ where λ is the wavelength beign observed, for our 850GHz receiver this limits the error to be 22μm RMS as max.

Each panel has 5 micrometer adjustment screws that can be used to re-position it, in figure 2 there is a diagram of one of the APEX panels. The 5 degrees of freedom in a panel can correct piston

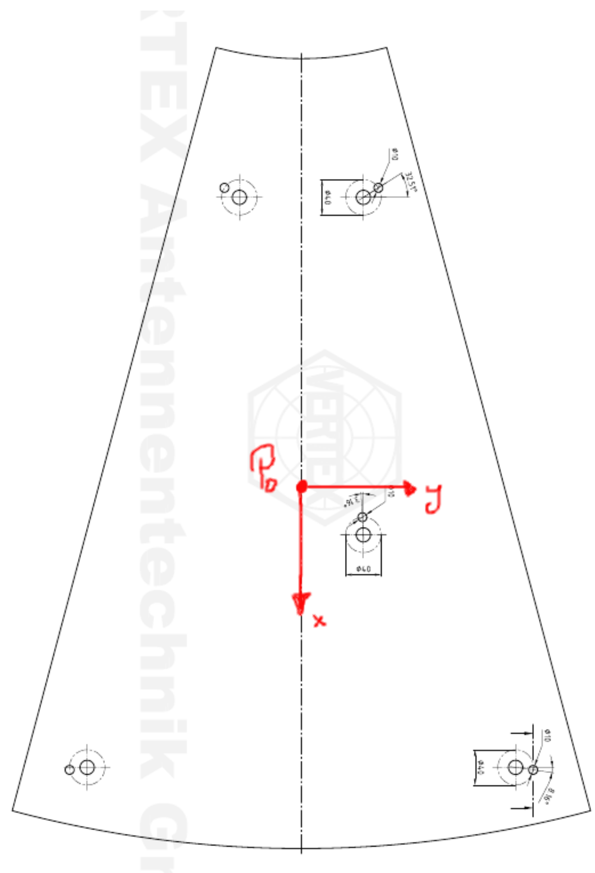

Figure 2: APEX panel example.

(if the panel has a global positive/negative offset), tilts and first order mechanical torsions. These type of deformations can be modeled as equation 1 where (x,y) are the local coordinates from each panel shown in the figure 2(more on that later..).

$$f_{\text{Panel}_i}(x, y) = a_i + b_i \cdot x + c_i \cdot y + d_i \cdot (x^2 + y^3) + e_i \cdot (x^2 - y^2) \tag{1}$$

To measure the surface of the antenna we make a map of its beam pattern. The beam pattern of one antenna can be roughly defined as the way that one antenna emits/receive power for different positions. So for example the desired antenna for radioastronomy is one that only sense a small region in the sky with a huge gain, therefore you can direct it to one astronomical source and be sure that you are just looking that.

In the figure 3 there is an example of the beam pattern measured at APEX at 92GHz, where you can see that most of the power is beign captured/emited in a small angular region near the 0°. To measure the beam pattern of the antenna you place a transmitter at some given distance and then start to move the antenna around it to get the received power in different positions.

The beam pattern of one antenna is related to the induced currents in the main dish, and therefore

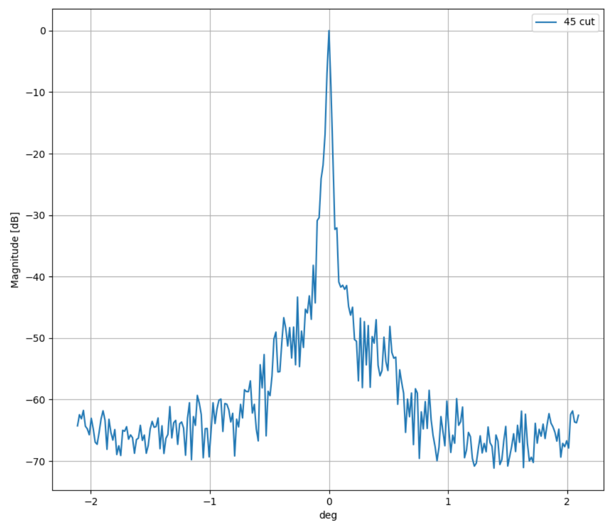

Figure 3: APEX beam pattern cut.

to the deviations of the position of the panels. As always, the actual expressions are awfull, but with some approximations you can find that the relation between the beam pattern and the currents is just a 2D Fourier transform + some corrections that you need to do over the data.

To truly invert a Fourier transform you need to measure amplitude and phase, then an holography system usually measure the complex beam pattern and got maps like the following at figure 4. After the processing we got the surface error maps shown at the left in figure 5.

With the final surface error we iterate over the panels area doing a fitting with the deformation function at equation 1 [1]. and finally we can compute the turns that we need to give to the screw adjuster of each panel.

[1]Note that here there is a subtle trick: since in the standard method we project the errors to a plane, then in this case (x,y) are not curved and can be taken as the usual cartesian coordinates

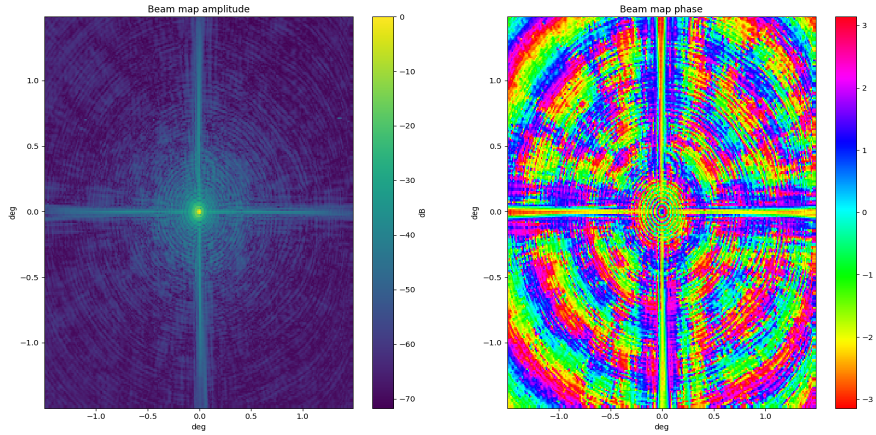

Figure 4: APEX holography map.

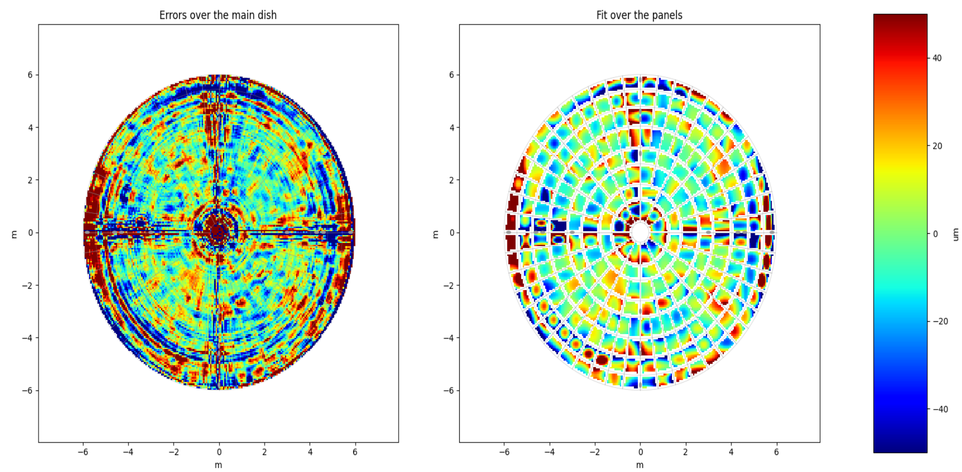

Figure 5: Derived surface error and the fit over the panels.

2 Antenna simulation

There are some problems with the standard method, so we want to look for a different way to do it. To start we want to simulate one antenna, that means having a given geometry we want to compute its beam pattern. For that you have the Maxwell equations, but solve them for big antennas at high frequencies will take ages, so we make some compromises.

The idea to simulate the antenna is to start from a known source, illuminate the geometry and compute the reflected field. The Kirchhoff-Fresnel equation shown in 2 is a scalar approximation to compute the reflected field that hits a given geometry. The beuty of this equation is that the incoming wave is reflected in a geometrical way, but each point generates a spherical wave and then you integrate everything. This naturally depicts the diffraction effects that we have in our measurement system.

$$E_r(\vec{r}) = \frac{-ik}{4\pi} \int E_i(\vec{r'}) \frac{e^{ik(|\vec{r}-\vec{r'}|}}{|\vec{r}-\vec{r'}|} (\hat{r}\cdot\hat{n})\, dS \tag{2}$$

So the main idea is:

1. We start from a horn antenna located in the optical path, this antenna emits a know field that we propagate all the way up to the subreflector (that have an hiperboloid geometry)

2. At the subreflector, we compute the reflected field and propagate it to the primary dish (that ideally has a parabolic geometry).

3. At the primary, we compute the reflected fields at 1835mts in an area of $3^\circ \times 3^\circ$ to have the beam map.

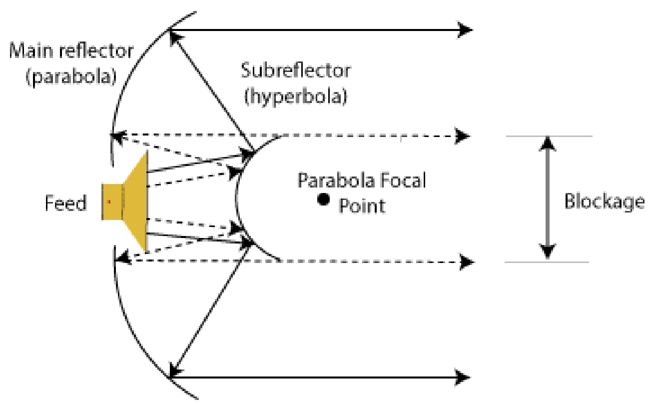

Figure 6: Diagram of the simulation

3 Geometry

Due the problem symetries we take cylindrical parametrization for all the geometries. Also is good to remember that we are dealing with curved surfaces, so the good orthogonal coordinates rely in the tangent vectors of the surface. In our case those are given by equations 3, with those we can compute the normal vectors as shown in 4 and the differential surface as 5.

$$\vec{t_r} = \frac{\partial\vec{r}}{\partial r} \qquad \vec{t_\phi} = \frac{\partial\vec{r}}{\partial\phi} \tag{3}$$

$$\hat{n} = \frac{\vec{t_r}\times\vec{t_\phi}}{|\vec{t_r}\times\vec{t_\phi}|} \tag{4}$$

$$dS = \vec{t_r}\times\vec{t_\phi} dr d\phi \tag{5}$$

So now we need to compute all these for the different geometries that we have in the problem.

3.1 Ideal main dish

Here we will start with the ideal paraboloid and then introduce the deformation over it. The ideal paraboloid is given by the equation 6 where f is the foci of the parabola.

$$\vec{S_0}(r,\phi) = \left(r\cos(\phi), r\sin(\phi), \frac{r^2}{4f}\right) \tag{6}$$

We build the curved coordinate system with the normalized tangent vectors, they are by construction an orthonormal basis (we will use them later)

$$\hat{e}_r = \left(\cos(\phi), \sin(\phi), \frac{r}{2f}\right)\frac{1}{\sqrt{1+\frac{r^2}{4f^2}}} \tag{7}$$

$$\hat{e}_\phi = (-\sin(\phi), \cos(\phi), 0) \tag{8}$$

The normal vectors are given by

$$\hat{n}_0 = \hat{e}_r \times \hat{e}_\phi = \left(-\frac{r}{2f}\cos(\phi), -\frac{r}{2f}\sin(\phi), 1\right) \cdot \frac{1}{\sqrt{1+\frac{r^2}{4f^2}}} \tag{9}$$

And the differential surface is given by

$$dS = r\sqrt{1+\frac{r^2}{4f^2}}drd\phi \tag{10}$$

3.2 Deformed main dish

I just want to start saying that all the expressions here are horrible, but the main idea is simple: we add a deformation pointing in the normal direction of the paraboloid surface. Since each panel is independent we create a local coordinate system for each panel based in the tangent vectors of the ideal paraboloid and we add the deformations there. The thing is that we need to compute once again the new tanget vectors, normal vectors and the differential surface of the modified paraboloid, which are once again super awfull.

For each panel in the dish we will generate a local coordinate system.

Once again we return to the panel example image at figure 7, now we will use (x, y) as as the tangent vectors coordinate system as the equations 11 and 12, where $\vec{p_{0i}}$ is the center of the panel i and $\vec{p_i}$ are the points that belong to that panel.

$$\vec{x_i'} = (\vec{p_i} - \vec{p_{0i}}) \cdot \hat{e}_r \tag{11}$$

$$\vec{y_i'} = (\vec{p_i} - \vec{p_{0i}}) \cdot \hat{e}_\phi \tag{12}$$

With that we can define the modified surface as 13 where we are adding deformations in the normal direction of the panels with the function $f(x, y, params)$ defined in equation 14 where $\text{params}_i =$

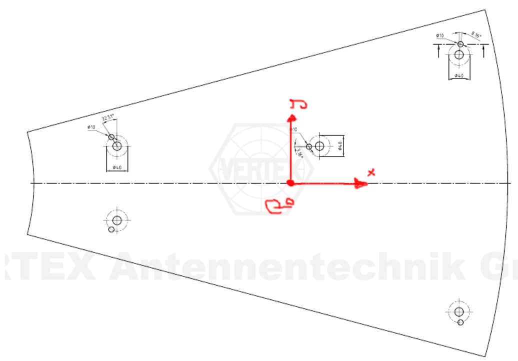

Figure 7: APEX panel example.

$\{a_i, b_i, c_i, d_i, e_i\}$ are the parameters that define the deformation of the panel i. The δ in the equation 13 is a Kronecker delta that limits the area of the panel i.

$$\vec{r}_{prim} = \vec{S_0}(r,\phi) + \sum_i \left[f(x'_i, y'_i, \text{params}_i) \cdot \hat{n}_0(r,\phi)\delta(r_{min,i} < r < r_{max,i}; \phi_{min,i} < \phi < \phi_{max,i}) \right] \tag{13}$$

$$f(x'_i, y'_i, \text{params}_i) = a_i + b_i x'_i + c_i y'_i + d_i({x'}^2_i + {y'}^2_i) + e_i({x'}^2_i - {y'}^2_i) \tag{14}$$

Now we once again need to compute the tangent and normal vectors to get the differential surface of this new surface.

After some tedious math we arrive to the following representation of the tangent vectors

$$\partial_r \vec{r}_{prim} = \partial_r S_0 + \frac{\partial f}{\partial x'} \left(\sqrt{1 + \frac{r^2}{4f^2}} + (\vec{p} - \vec{p}_0) \cdot \frac{\partial \hat{e}_r}{\partial r} \right) \hat{n}_0 + f(x', y') \frac{\partial \hat{n}_0}{\partial r} \tag{15}$$

$$\partial_\phi \vec{r}_{prim} = \partial_\phi S_0 + \left[(\vec{p} - \vec{p}_0) \cdot \left(\frac{\partial f}{\partial x'} \frac{\partial \hat{e}_r}{\partial \phi} + \frac{\partial f}{\partial y'} \frac{\partial \hat{e}_\phi}{\partial \phi} \right) + r \frac{\partial f}{\partial y'} \right] \hat{n}_0 + f(x', y') \frac{\partial \hat{n}_0}{\partial \phi} \tag{16}$$

And finally we compute the normal vectors as

$$\vec{n} = \partial_r \vec{r}_{prim} \times \partial_\phi \vec{r}_{prim} \tag{17}$$

And build the differential surface as:

$$dS_{prim} = |\partial_r \vec{r}_{prim} \times \partial_\phi \vec{r}_{prim}| dr d\phi \tag{18}$$

3.3 Hyperboloid

The subreflector positions are constructed with the equation 19 where the hyperboloid parameters are a, b.

$$\vec{r}_{sec} = \left(r\cos(\theta), r\sin(\theta), a\sqrt{\frac{r^2}{b^2-1}} - a\right) \tag{19}$$

The normal vectors are given by 21 and the differential area by equation 22.

$$\partial z_r = ar/(b^2\sqrt{r^2/b^2+1}) \tag{20}$$

$$\vec{n}_{sec} = \big(-\partial z_r\cos(\phi), -\partial z_r\sin(\phi), 1\big)\frac{1}{\sqrt{1+(\partial z_r)^2}} \tag{21}$$

$$dS_{sec} = r\sqrt{1+(\partial z_r)^2} \tag{22}$$

3.4 Horn beam

Located in the optical path we have a horn antenna, which pattern can be represented by the gaussian beam equations at 27. These equations are written in cylindricals, with z pointing outwards the horn antenna. The figure 8 shows the beam propagation of the gaussian beam. Each horn has a parameter ω_0 that is called beam waist, this can be obtained from the geometry of the horn.

The equations look awfull, but if we keep the subreflector and horn at fix postion, we just need to evaluate the function 27 at the postions $\vec{r}_{sec}$ a single time and then we can reuse it.

$$z_c = \frac{\pi\omega_0^2}{\lambda} \tag{23}$$

$$\omega(z) = \omega_0\left[1+\left(\frac{z}{z_c}\right)^2\right]^{0.5} \tag{24}$$

$$R(z) = z + \frac{z_c^2}{z} \tag{25}$$

$$\phi(z) = tan^{-1}(z/z_c) \tag{26}$$

$$E_{horn}(r,z) = \sqrt{\frac{2}{\pi\omega(z)^2}}exp\left(-\frac{r^2}{\omega(z)^2} - ikz - \frac{i\pi r^2}{\lambda R(z)} + i\phi(z)\right) \tag{27}$$

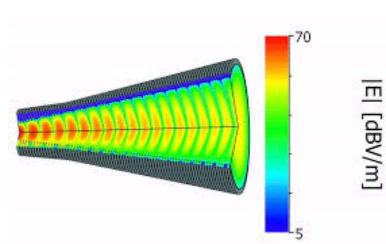

Figure 8: Gaussian beam from a horn.

4 Summary of the antenna simulation

There are plenty of awfull equations, but in principle we just need to evaluated them in the proper values

- If we left the horn and the subreflector at fixed position, then we only need to compute a single time $\vec{r}_{sec}, \hat{n}_{sec}, E_{horn}(\vec{r}_{sec}), dS_{sec}$.
- When deforming the primary we will need to update $\vec{r}_{prim}, \hat{n}_{prim}, dS_{prim}$ for the sets $\{a_i, b_i, c_i, d_i, e_i\} i \in \{1, .., 265\}$
- We compute the reflected field of the secondary to the primary positions, for this integral the only thing that may vary between iterations are the $\vec{r}_{prim}$.

$$E_{prim}(\vec{r}_{prim}) = \frac{-ik}{4\pi} \int_{r' \in \vec{r}_{sec}} E_{horn}(\vec{r'}) \frac{e^{ik(|\vec{r}_{prim} - \vec{r'}|}}{|\vec{r}_{prim} - \vec{r'}|} (\hat{r}_{prim} \cdot \hat{n}_{sec}) \, dS_{sec} \tag{28}$$

- We compute the reflected field of the primary to the final positions with the equations.

$$E_{final}(\vec{r}_{final}) = \frac{-ik}{4\pi} \int_{r' \in \vec{r}_{prim}} E_{horn}(\vec{r'}) \frac{e^{ik(|\vec{r}_{final} - \vec{r'}|}}{|\vec{r}_{final} - \vec{r'}|} (\hat{r}_{final} \cdot \hat{n}_{prim}) \, dS_{prim} \tag{29}$$

5 Optimization

The final idea of all this development is to introduce a measured beam map and to tell the optimizer ”find the set of deformation parameters that generates the most similar beam map to the measured one”.

For that we used an adam optimizer with the coefficients $\{a_i, b_i, c_i, d_i, e_i\} i \in \{1, .., 265\}$ as parameters to tune (1325 parameters in total) and we use the loss function as the expression in 30 where the regularization parameter is to penalize the surface RMS added via the deformation.

$$Loss = \sum_{r \in \vec{r}_{final}} (E_{final}(r) - E_{measured}(r))^2 + \lambda \underbrace{\sum_i (f(x', y', \text{params}_i))^2}_{regularization} \tag{30}$$

We made all the code using the Jax library, then we can rely in the automatic differentiation to make partial derivatives for the deformation coefficients allowing to use gradient descendent optimization.

6 Machine learning digression

Being practical, what we really want is to found a map between beam map $\rightarrow$ panel deformation, The antenna simulation gave us the map: panel deformation $\rightarrow$ beam pattern, but there is no clear way to go the other way, therefore we used the optimizer to fit the map, tunning the paramerters by trial and error until we got the most similar beam pattern.

Another approach is use these machine learning-like algorithms to obtain the mapping from beam map to panel deformations right away. Since we have the antenna simulation path, we have the option to generate a dataset with any size to train a model to output the deformations that produced a given beam map. The biggest advantage of taking this approach is that all the math complexity will be hide and if we are lucky the dimensionality of space of parameters can be lower.

I dont know if that is better suited for the annelear, where we give samples of an input field and the output field, and optimize some sort of model to do the mapping between them.